

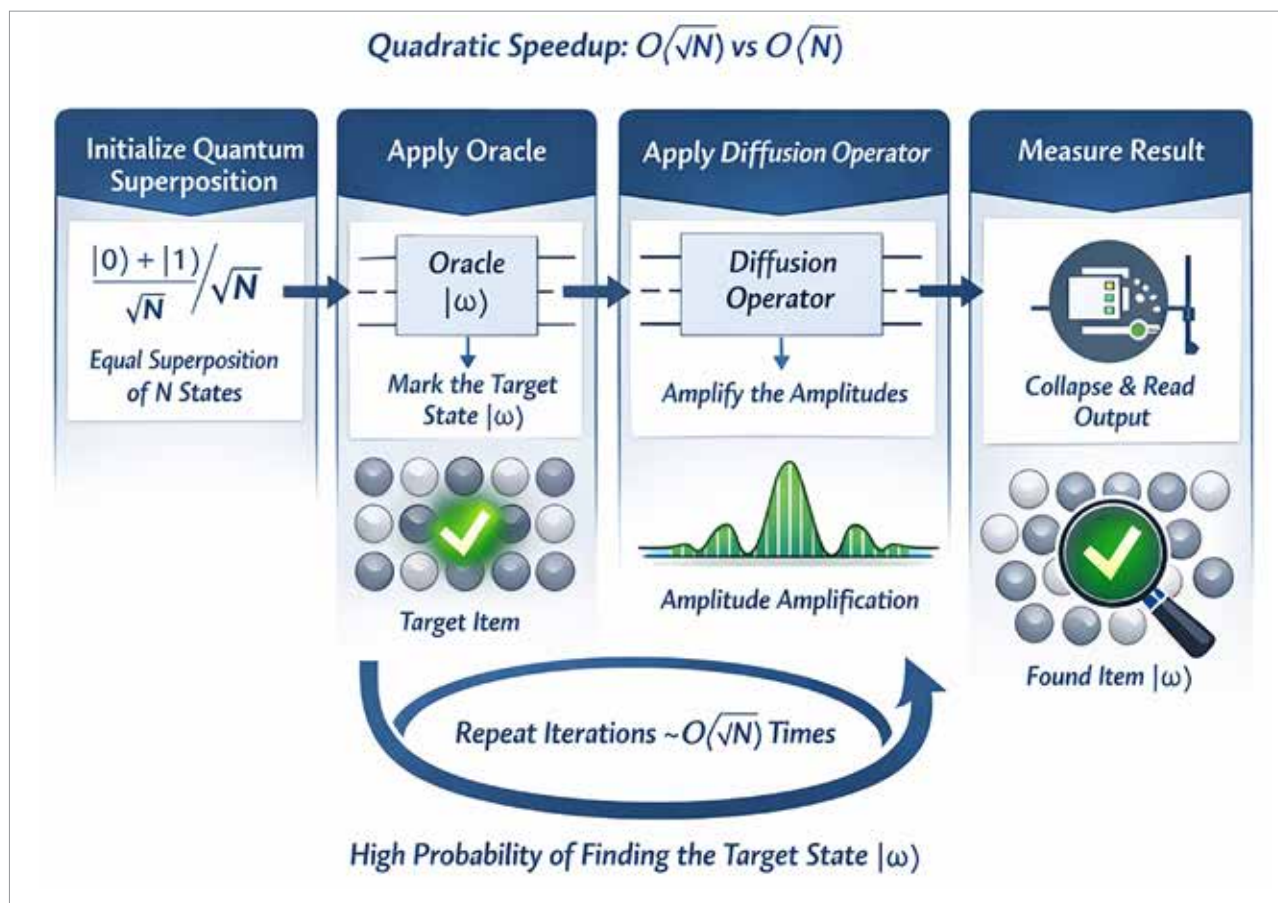


Figure 2: Grover's quantum search algorithm

algorithms, such as the Quantum Approximate Optimisation Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE), are designed to work with noisy intermediate-scale quantum (NISQ) devices. These approaches combine quantum circuits with classical optimisation routines, enabling practical applications despite hardware limitations.

Recent work includes quantum algorithms for simulating chemical reactions, solving linear systems (Harrow-Hassidim-Lloyd algorithm), and quantum walks for graph analysis. These advances are expanding the scope of quantum computing beyond traditional factoring and search tasks.

Design approach: Hybrid algorithms typically involve parameterised quantum circuits, where classical computers optimise the parameters to minimise or maximise a cost function.

```
# Example: Variational Quantum Eigensolver (VQE)
from qiskit.algorithms import VQE
from qiskit.circuit.library import TwoLocal
ansatz = TwoLocal(rotation_blocks='ry', entanglement_
blocks='cz')
vqe = VQE(ansatz)
```

Quantum machine learning: Algorithms for AI

Quantum machine learning (QML) is an emerging field that explores how quantum algorithms can enhance artificial intelligence. Algorithms such as Quantum Support Vector Machines, Quantum Principal Component Analysis, and Quantum Neural Networks promise to speed up data processing and pattern recognition. Variational quantum algorithms, in particular, are well-suited for NISQ devices and leverage quantum circuits to model complex data distributions.

Example use case: Quantum classifiers can analyse large datasets more efficiently, potentially transforming areas like fraud detection and image recognition.

The code snippet is:

```
# Quantum Support Vector Machine (QSVM) skeleton
from qiskit_machine_learning.algorithms import QSVM
qsvm = QSVM(training_dataset, test_dataset)
```

While practical quantum advantage for machine learning remains an open question, ongoing research is bridging the gap between theory and real-world application.